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BENGALURU, INDIA

WHERE THE

WORLD COMES

TO LEARN

TRADITIONAL MEDICINE AND HEALTH MANAGEMENT- BBT755C

Dr.Ashalatha

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Department of Biotechnology



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Overveiw

Acharya stands as a beacon of excellence in higher education, boasting a legacy of academic distinction since its establishment in 1990. We offer a transformative educational experience, fostering holistic development, nurturing innovation and providing world-class facilities to ensure an enriching journey for our students.

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MODULE – 2 (8 hours) CLINICAL SCIENCES

Causes and mechanisms of disease, describing diseases, the principles of differential diagnosis. Disorders of cells and response to disorders. Infectious diseases and circulatory disorders. Symptoms and signs related to diseases of the various body systems and diseases of control systems.

Causes of Disease (Etiology)

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- **Genetic Causes**
- **Definition:** Diseases caused by abnormalities in DNA or chromosomes.
- **Mechanism:** Mutations alter protein structure/function → defective enzymes or structural proteins.
- **Examples:**
  - *Single gene mutation:* Sickle cell anemia (abnormal hemoglobin).
  - *Chromosomal abnormality:* Down syndrome (extra chromosome 21).
  - *Inherited disorders:* Cystic fibrosis (defective chloride channel).

Causes of Disease (Etiology)

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- **Environmental Causes**
- **Physical agents:**
  - Trauma → fractures, wounds.
  - Radiation → DNA mutations, cancers.
  - Temperature extremes → burns, frostbite.
- **Chemical agents:**
  - Poisons (arsenic, lead) damage tissues.
  - Drugs/alcohol cause liver/kidney damage.
- **Biological agents:**
  - Bacteria (e.g., TB, cholera).
  - Viruses (HIV, influenza).
  - Fungi (ringworm).
  - Parasites (malaria).

Causes of Disease (Etiology)

- **Nutritional Causes**
- **Deficiency:** Lack of vitamins/minerals/proteins.
  - Vitamin D deficiency → rickets.
  - Vitamin C deficiency → scurvy.
  - Iron deficiency → anemia.
- **Excess:**
  - Overeating → obesity, diabetes.
  - Hypervitaminosis A/D → toxicity
- **Immunological Causes**
- **Hypersensitivity:** Overreaction of the immune system to harmless substances.
  - Example: Asthma, hay fever.
- **Autoimmunity:** Immune system attacks self-tissues.
  - Example: Rheumatoid arthritis, lupus.
- **Immunodeficiency:** Weak/absent immune system → frequent infections.
  - Example: HIV/AIDS, congenital immunodeficiencies.

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Causes of Disease (Etiology)

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- **Lifestyle Causes**
  - **Smoking:** Increases risk of lung cancer, COPD, heart disease.
  - **Alcohol:** Liver cirrhosis, brain damage, cancer.
  - **Poor diet/lack of exercise:** Obesity, diabetes, hypertension.
  - **Stress:** Increases risk of depression, heart disease, ulcers.
- **Aging**
  - Natural degeneration of cells/tissues with age.
  - Reduced repair capacity, increased oxidative stress.
  - Examples: Osteoporosis, Alzheimer's disease, Parkinson's disease.

Mechanism of diseases

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- Pathology, the scientific study of disease, is the objective description of the traits, causes, and effects of abnormal conditions. Pathologic conditions involve measurable changes in normal structure and function that threaten the internal stability, or **homeostasis,** of the body.
- A defined collection of signs and symptoms that characterize a disorder or condition is termed a *syndrome*.
- The development of disease occurs in stages, described as the **pathogenesis.** In the course of infection, for instance, the pathogenesis may include an incubation period, a period of full-blown symptoms, and then remission or convalescence. The pathogenesis of a disease varies with the individual patient, the causative factors, and medical intervention.

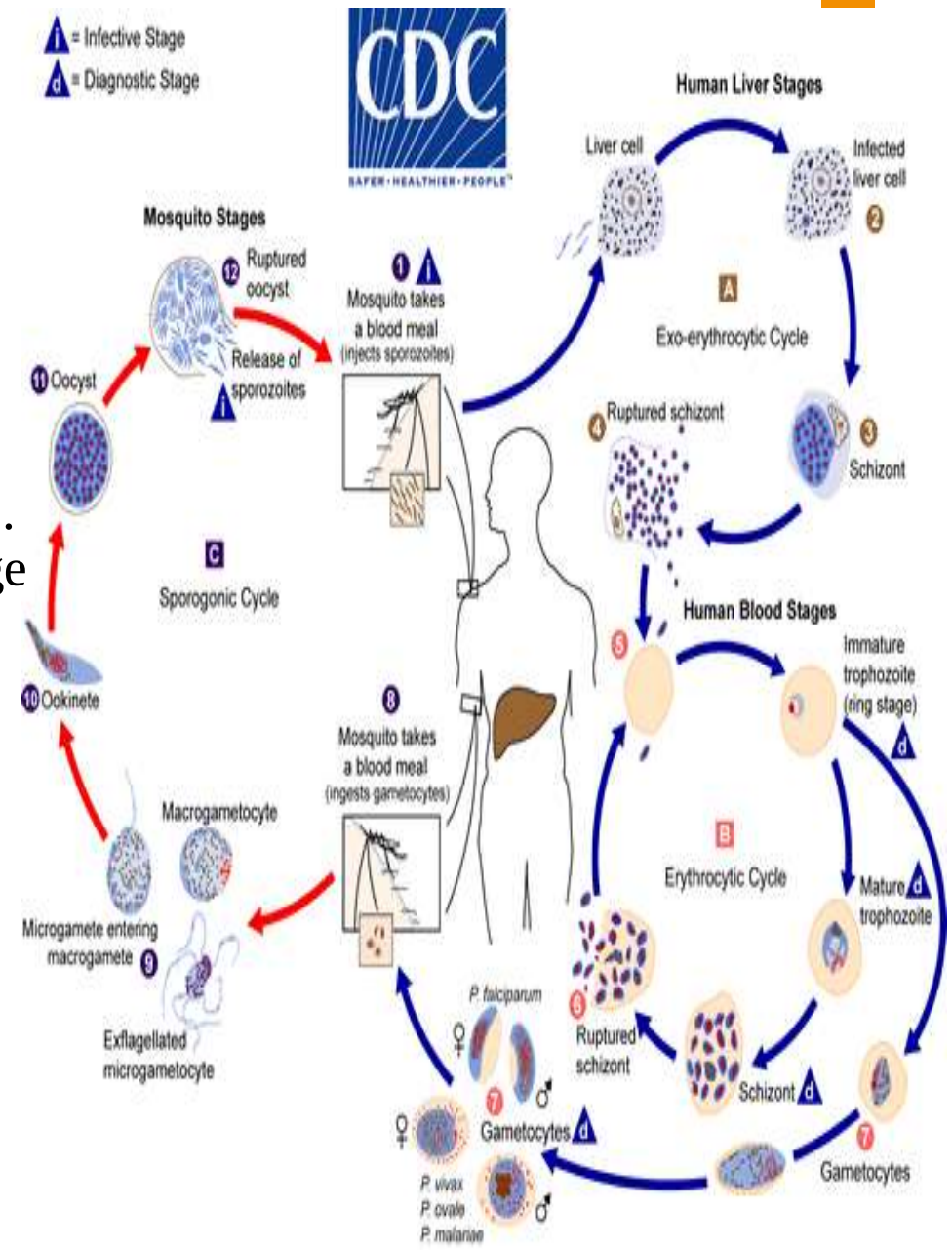
Mechanism of diseases

- **Genetic Mechanisms**
- **Definition:** Diseases caused by defects in DNA or chromosomes.
- **How it works:**
  - Mutation in genes → abnormal protein or enzyme.
  - Cells lose normal function or gain harmful function.
  - Leads to structural/functional abnormalities in tissues.
- **Examples :**
  - *Single-gene disorders*: Sickle cell anemia (defective hemoglobin), Cystic fibrosis (CFTR mutation).
  - *Chromosomal disorders*: Down syndrome (extra chromosome 21), Turner syndrome (missing X chromosome).
  - *Multifactorial*: Diabetes, hypertension, cancer susceptibility.

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Mechanism of diseases

- **Infectious Mechanisms**
- **Definition:** Diseases caused by microorganisms (bacteria, viruses, fungi, protozoa, parasites).
- **Steps of Pathogenesis:**
  - Entry into host (inhalation, ingestion, vector bite, direct contact).
  - Multiplication and colonization in tissues.
  - Production of toxins/enzymes that damage cells.
  - Immune system reaction → fever, inflammation, pus, etc.
- **Examples:**
  - Tuberculosis (bacteria damaging lungs).
  - HIV/AIDS (virus destroying immune cells).
  - Malaria (parasite destroying red blood cells).



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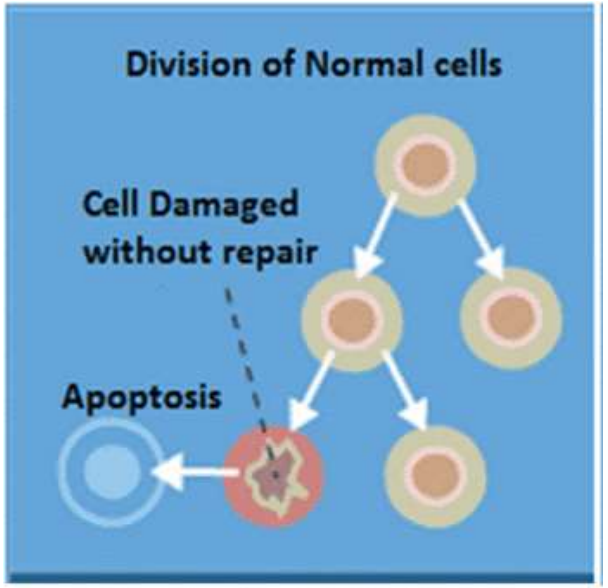
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ORGANISM	RESERVOIR	INFECTION OR DISEASE
BACTERIA		
Escherichia coli (E. coli)	Colon, manure	Enteritis, mild to severe
Staphylococcus aureus	Skin, hair, anterior nares	Wound infection, pneumonia, food poisoning, cellulitis
Streptococcus (β-hemolytic group A) organisms	Oropharynx, skin, perianal area	"Strep throat," rheumatic fever, scarlet fever, impetigo
Streptococcus (β-hemolytic group B) organisms	Adult genitalia	Urinary tract infection, wound infection, endometritis
Mycobacterium tuberculosis	Lungs	Tuberculosis
Neisseria gonorrhoeae	Genitourinary tract, rectum, mouth, eye	Gonorrhea, pelvic inflammatory disease, infectious arthritis, conjunctivitis
Rickettsia rickettsii	Wood tick	Rocky Mountain spotted fever
Staphylococcus epidermidis	Skin	Wound infection, bacteremia
VIRUSES		
Hepatitis A virus	Feces, blood, urine	Hepatitis A (infectious hepatitis)
Hepatitis B virus	Feces, blood, all body fluids and excretions	Hepatitis B (serum hepatitis)
Hepatitis C virus	Blood and body fluids	Liver disease may become chronic
Herpes simplex virus	Lesions of mouth, skin, blood, excretions	Cold sores, aseptic meningitis, sexually transmitted disease
Human immunodeficiency virus (HIV)	Blood, semen, vaginal secretions (also isolated in saliva, tears, urine, breast milk, but not proven to be sources of transmission)	Acquired immunodeficiency syndrome (AIDS)
Hantavirus	Deer mouse urine, feces, and saliva	Upper respiratory infection (URI) to lower respiratory infection (LRI) to adult respiratory distress syndrome (ARDS)
West Nile virus	Mosquito-borne	Fever, rash, hepatitis, encephalitis
FUNGI		
Aspergillus organisms	Soil, dust	Aspergillosis
Candida albicans	Mouth, skin, colon, genital tract	Thrush, dermatitis
PROTOZOA		
Plasmodium falciparum	Mosquito	Malaria

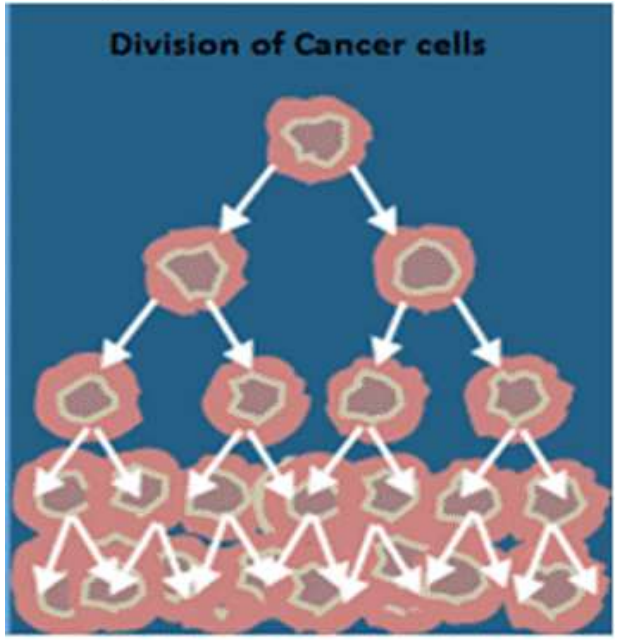
Common pathogens and some infections or diseases that they cause

Neoplastic Mechanisms

- **Definition:** Diseases due to uncontrolled cell growth forming tumors.
- **How it happens:**
  - Mutation in cell cycle control genes (oncogenes, tumor suppressor genes).
  - Uncontrolled proliferation → tumor formation.
  - Tumors can be:
    - **Benign:** Localized, slow growth.
    - **Malignant (cancer):** Invades tissues, spreads (metastasis).
- **Examples:**
  - Benign: Fibroid in uterus.
  - Malignant: Lung cancer, cervical cancer (HPV-related).



Apoptosis= cell death



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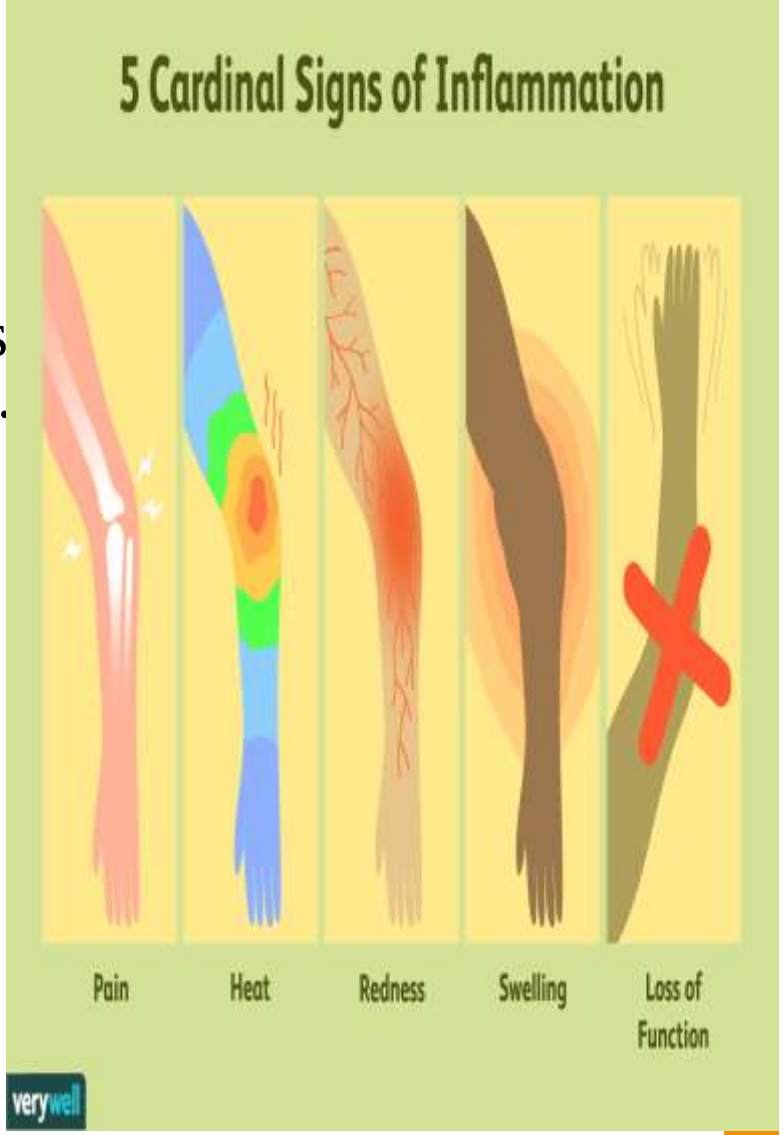
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Immune & Inflammatory Mechanisms

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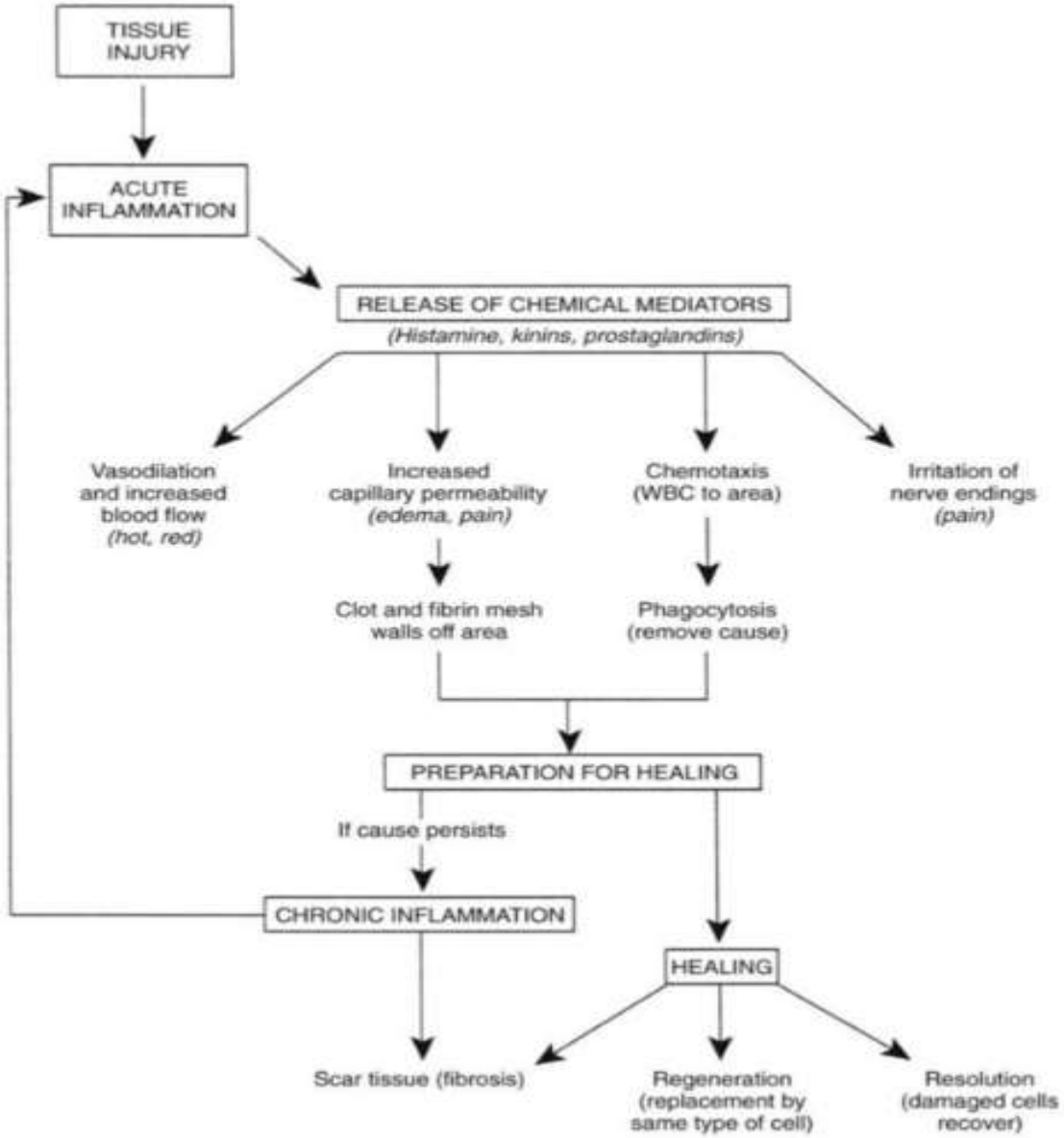
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- **Definition:** Diseases due to abnormal immune system activity.
- **Types:**
  - **Autoimmunity:** Immune system attacks own body (Rheumatoid arthritis, Lupus).
  - **Hypersensitivity:** Overreaction to harmless antigen (Asthma, Allergic rhinitis).
  - **Immunodeficiency:** Weak or absent immune system (HIV/AIDS).
- **Inflammation:**
  - Acute (short-term, protective) → e.g., sore throat, appendicitis.
  - Chronic (long-term, damaging) → e.g., Crohn's disease, chronic hepatitis.



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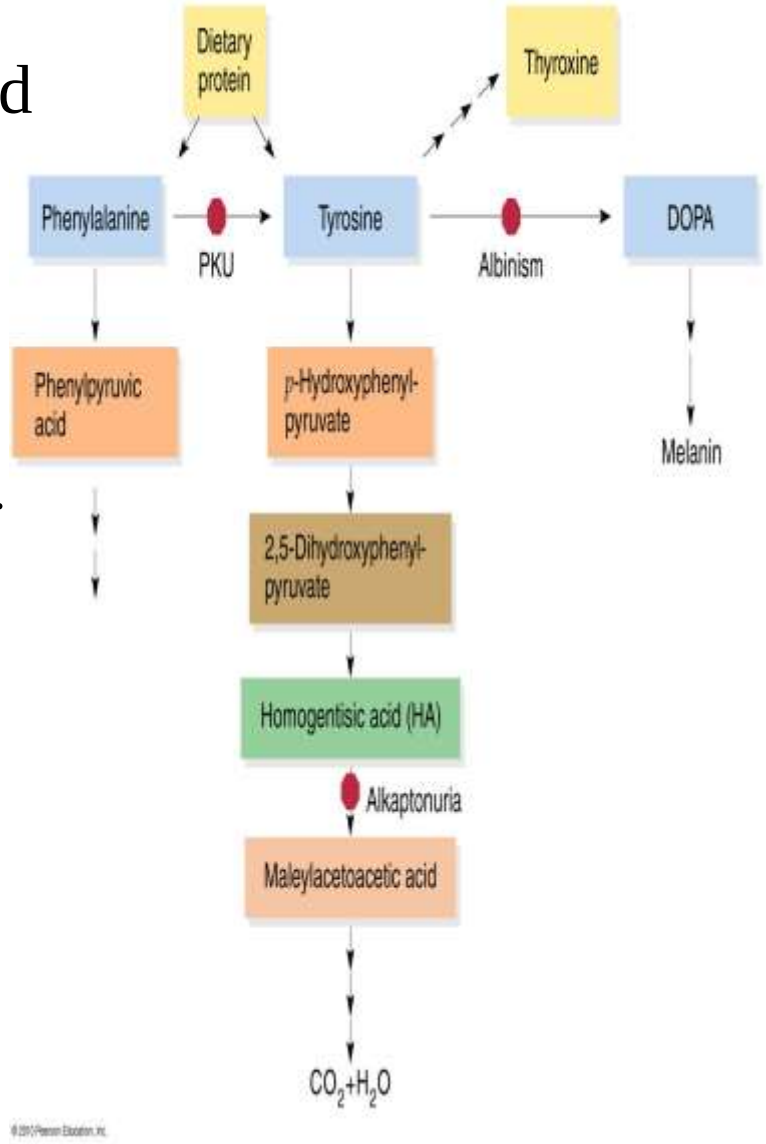
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The course of inflammation and healing. (From Gould BE: Pathophysiology for the health professions, ed 4, St Louis, 2011, Saunders.)

Metabolic Mechanisms

- **Definition:** Disorders caused by enzyme defects, hormonal imbalance, or disrupted biochemical pathways.
- **How it happens:**
  - Enzyme deficiency → buildup of toxic substances.
  - Hormonal imbalance → altered metabolism.
  - Nutrient processing fails → organ damage.
- **Examples:**
  - Diabetes mellitus (insulin deficiency or resistance → high blood sugar).
  - Gout (uric acid crystals in joints → inflammation).
  - Phenylketonuria (enzyme defect → brain damage if untreated)

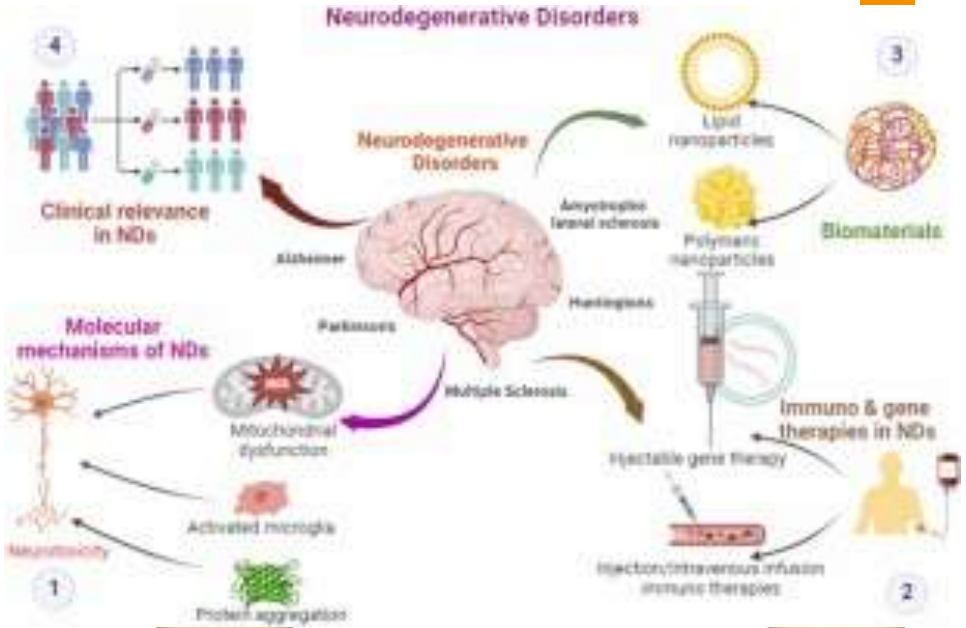


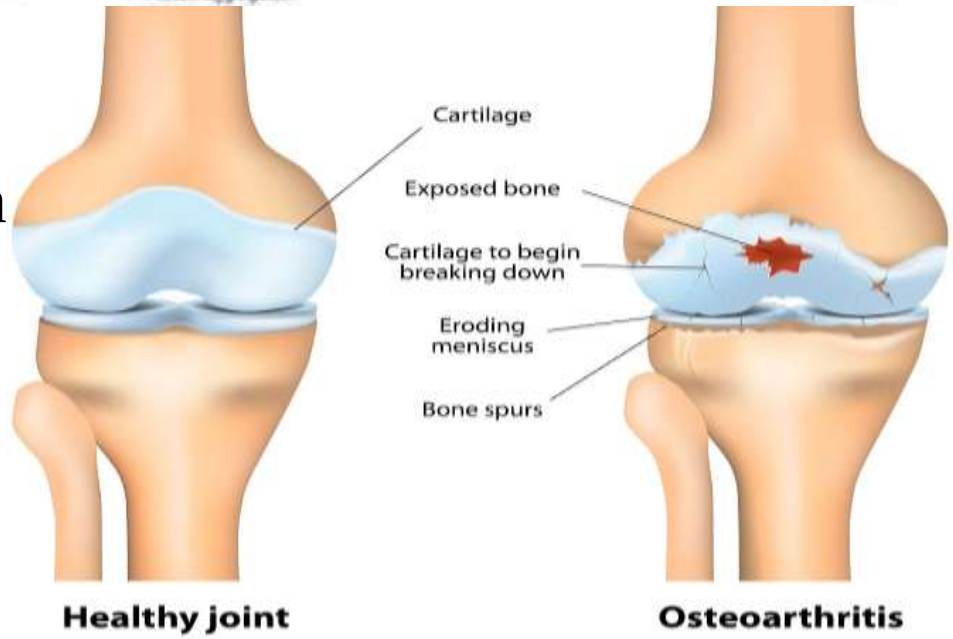
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Degenerative Mechanisms

- **Definition:** Diseases caused by gradual deterioration of cells and tissues, often age-related.
- **How it happens:**
  - Cellular wear and tear, oxidative stress, reduced repair.
  - Progressive cell loss → organ dysfunction.
- **Examples:**
  - Osteoarthritis (cartilage degeneration in joints).
  - Alzheimer's disease (loss of brain neurons, memory loss).
  - Parkinson's disease (loss of dopamine neurons).





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Describing Diseases

- → Definition
- → Cause
- → Mechanism
- → Clinical features
- → Complications
- → Diagnosis
- → Treatment
- → Prevention

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Disorders of cells and responses

- Cells are constantly exposed to stress, injury, and environmental changes.
- The way a cell responds depends on:
  - Type of cell (e.g., neurons vs. epithelial cells)
  - Type of injury (hypoxia, toxins, infection)
  - Duration (acute vs. chronic exposure)
  - Severity (mild vs. severe)
- Outcomes: Adaptation, Injury (reversible or irreversible), or Death.

Cell Injury

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- **Reversible cell injury**
  - Early, mild stage
  - Cell can return to normal if stimulus is removed
  - Features: cell swelling, fatty changes, membrane blebbing
- **Irreversible cell injury**
  - Severe, persistent damage
  - Leads to **cell death**
  - Features: mitochondrial damage, membrane rupture, nuclear changes

Cellular Adaptations (Response to Stress)

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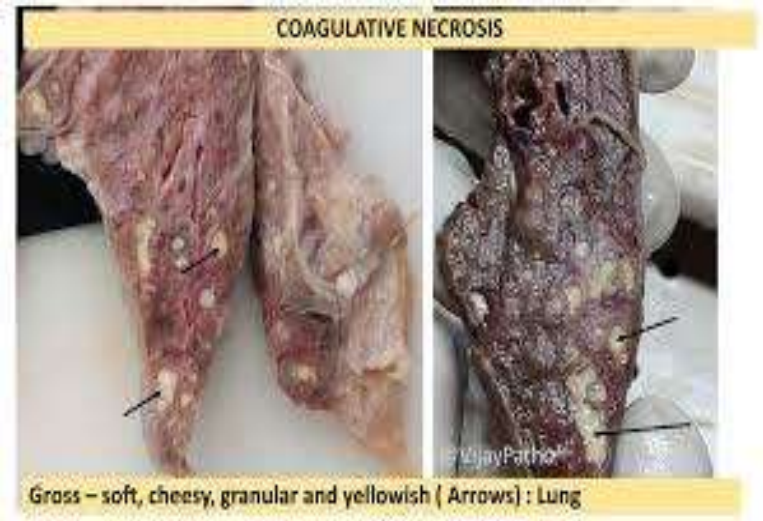
- When stressed, cells adapt to survive. Types include:
- **Atrophy** – Decrease in cell size (e.g., muscle wasting after immobilization)
- **Hypertrophy** – Increase in cell size (e.g., cardiac hypertrophy in hypertension)
- **Hyperplasia** – Increase in cell number (e.g., endometrial hyperplasia)
- **Metaplasia** – Replacement of one cell type by another (e.g., smoker's respiratory epithelium changing from columnar → squamous)
- **Dysplasia** – Abnormal cell growth and differentiation, often precancerous

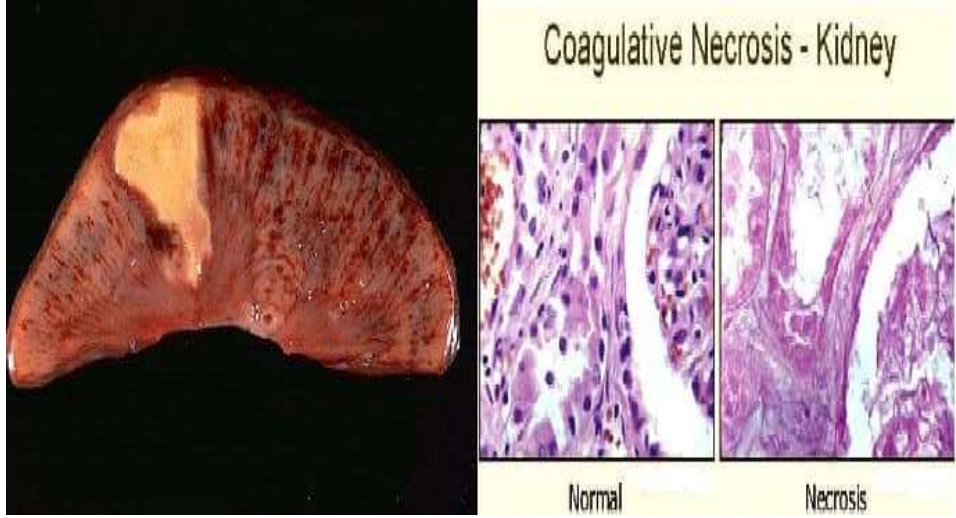
Cell Death

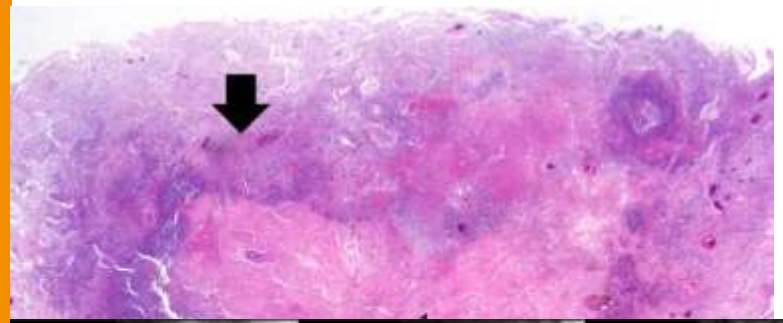
- **Necrosis (Uncontrolled, Pathological death)**
  - Always pathological
  - Types:
    - *Coagulative necrosis* (myocardial infarction)
    - *Liquefactive necrosis* (brain abscess, stroke)
    - *Caseous necrosis* (tuberculosis)
    - *Fat necrosis* (acute pancreatitis)
    - *Gangrenous necrosis* (limb ischemia)
- **Apoptosis (Programmed cell death)**
  - Controlled, physiological or pathological
    - Important in embryogenesis, immune regulation, eliminating damaged cells
  - Physiological roles:
    - Embryogenesis (removal of webbing in fingers)
    - Elimination of self-reactive lymphocytes
    - Removal of old/damaged cells
  - Pathological roles:
    - DNA damage from radiation/chemotherapy
    - Viral infections (e.g., hepatitis)
  - Features:
    - Cell shrinkage, chromatin condensation, apoptotic bodies
    - No inflammation

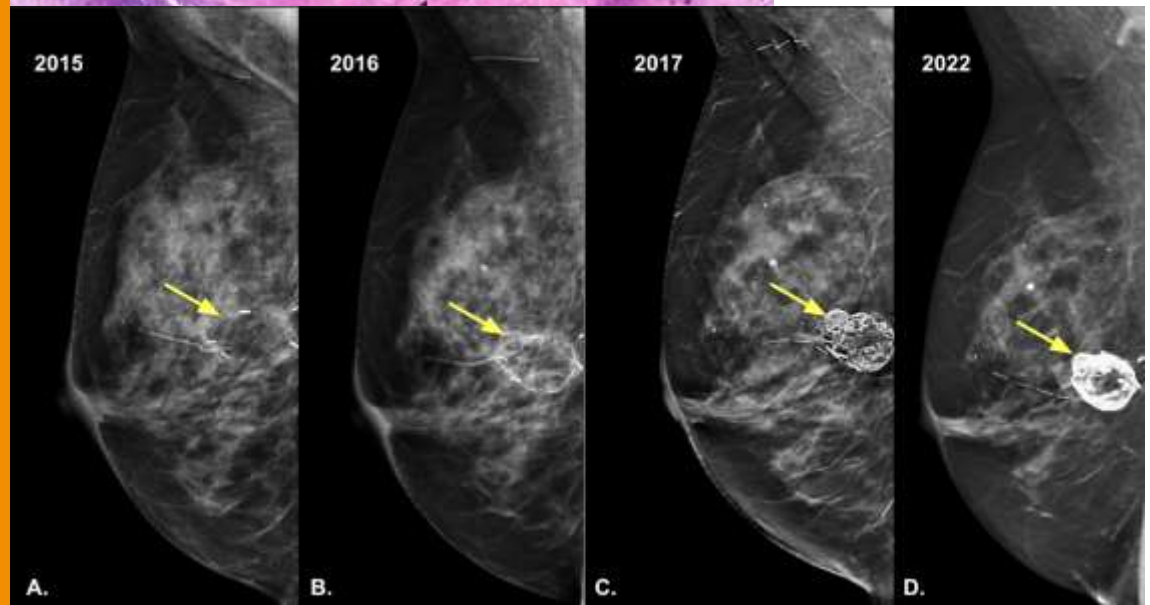
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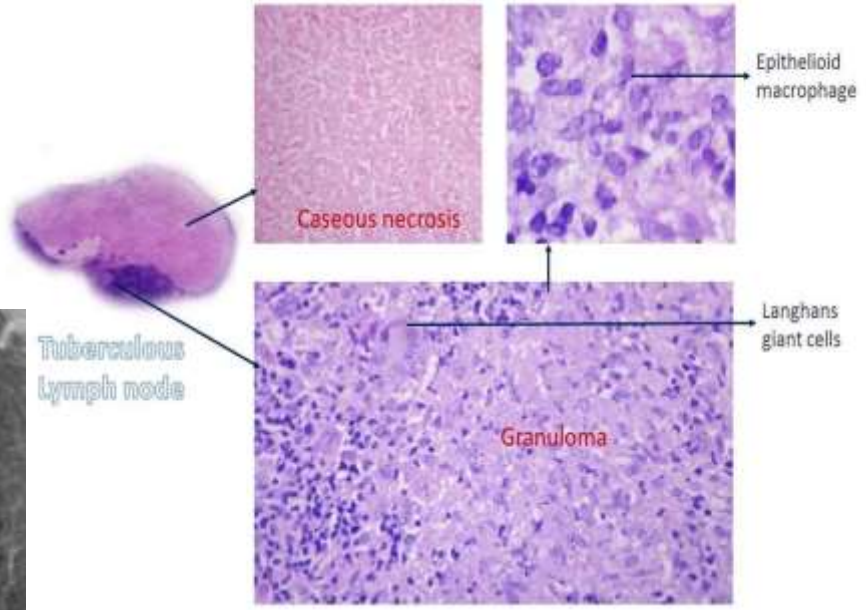
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- **Autophagy**
  - “Self-eating” process where cell digests its own organelles for survival during starvation
  - Plays role in cancer and degenerative disorders
- **Cellular Aging (Senescence)**
- Progressive decline in cell function with age
- **Mechanisms:**
  - Telomere shortening
  - Accumulation of DNA damage
  - Reduced repair and regenerative capacity
- **Clinical impact:** Alzheimer’s, Parkinson’s, osteoporosis

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INFECTIOUS DISEASES

INFECTIOUS DISEASES

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- Infectious diseases are illnesses caused by pathogenic microorganisms such as bacteria, viruses, fungi, and parasites.
- They spread directly (person-to-person) or indirectly (via food, water, air, vectors).
- Major global health concern → cause epidemics and pandemics (e.g., COVID-19, tuberculosis, malaria).

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Infectious Agents

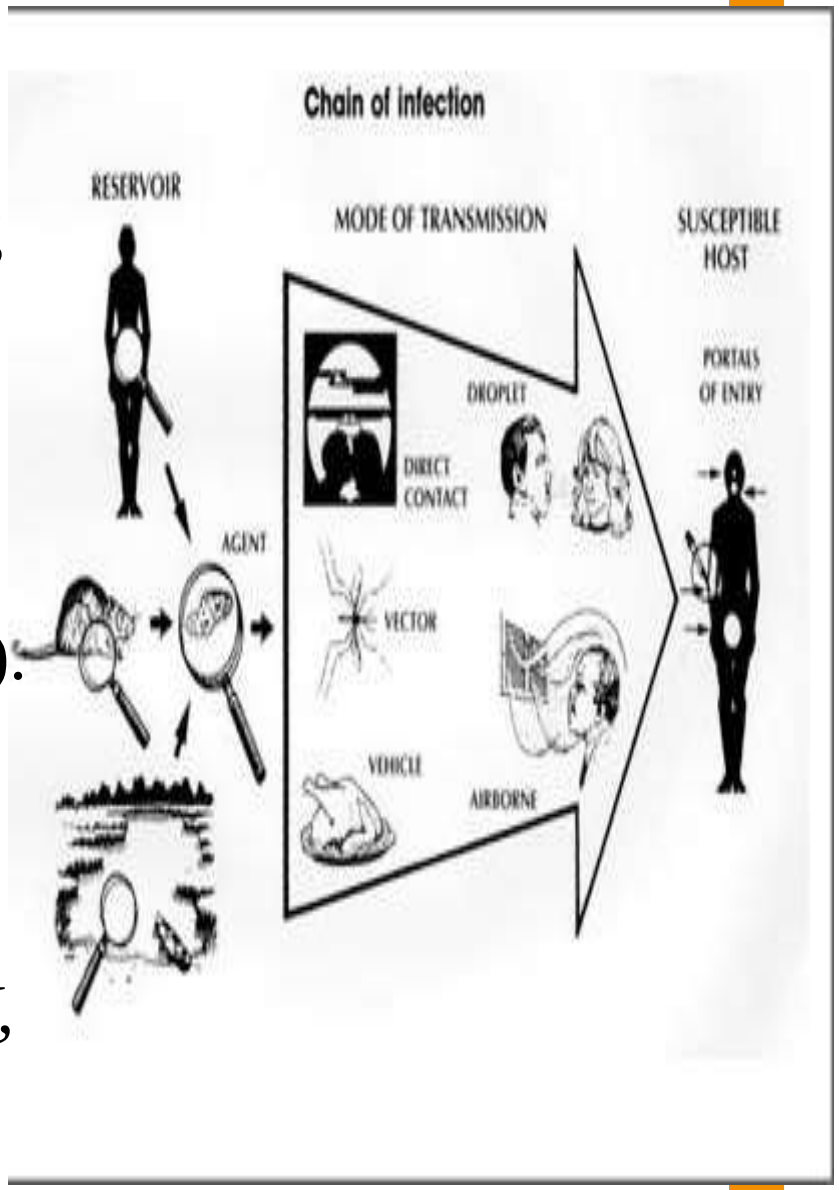
- **Bacteria**
  - Prokaryotic, reproduce by binary fission.
  - Can release **toxins**:
    - *Exotoxins* (secreted proteins; e.g., diphtheria toxin, tetanus toxin)
    - *Endotoxins* (lipopolysaccharide in Gram-negative bacteria causing septic shock)
  - Diseases: Tuberculosis, Typhoid, Pneumonia, Cholera.
- **Viruses**
  - Acellular, obligate intracellular parasites.
  - Destroy host cells by lysis or immune attack.
  - Diseases: Influenza, COVID-19, HIV, Hepatitis, Rabies.
- **Fungi**
  - Eukaryotic, can be unicellular (yeasts) or multicellular (molds).
  - Opportunistic in immunocompromised hosts.
  - Diseases: Candidiasis, Aspergillosis, Ringworm.
- **Parasites**
  - **Protozoa** (single-celled, e.g., *Plasmodium* → malaria)
  - **Helminths** (worms, e.g., Ascaris, Tapeworms)
  - **Ectoparasites** (ticks, lice, fleas).
- **Prions**
  - Infectious misfolded proteins, rare.
  - Diseases: Creutzfeldt–Jakob disease

Modes of Transmission

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- **Direct Contact:** skin-to-skin, sexual (HIV, syphilis, herpes).
- **Droplet/Airborne:** coughing, sneezing (TB, COVID-19, measles).
- **Fecal-Oral Route:** contaminated food/water (cholera, hepatitis A, polio).
- **Vector-Borne:** mosquito, tick, fly (malaria, dengue, chikungunya, plague).
- **Blood-borne:** transfusions, shared needles (HIV, hepatitis B & C).
- **Vertical (Mother → Child):** during pregnancy, birth, or breastfeeding (HIV, rubella, syphilis).



Pathogenesis – Steps in Infection

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Entry into host – via respiratory tract, GI tract, urogenital tract, skin breaks.

Adherence to host cells – surface adhesins, pili, capsules.

Colonization and multiplication – inside or outside cells.

Invasion & spread – through tissues, blood, lymph.

Evasion of immunity – antigenic variation, capsules, inhibiting phagocytosis.

Damage to host tissues

Direct cell killing	Toxin production	Inducing excessive immune response (e.g., cytokine storm in COVID-19).

Major Infectious Diseases

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A. Respiratory Infections	Gastrointestinal	C. Vector Borne Infections
• **Examples:** Tuberculosis, Influenza, COVID-19, Pneumonia, Whooping cough. • **Transmission:**	• **Examples:** Cholera, Typhoid, Hepatitis A & E, Amoebiasis, Rotavirus diarrhea. • **Transmission:**	• **Examples:** Malaria, Dengue, Zika, Chikunguny a, Yellow fever, Plague. • **Transmission:**

Major Infectious Diseases

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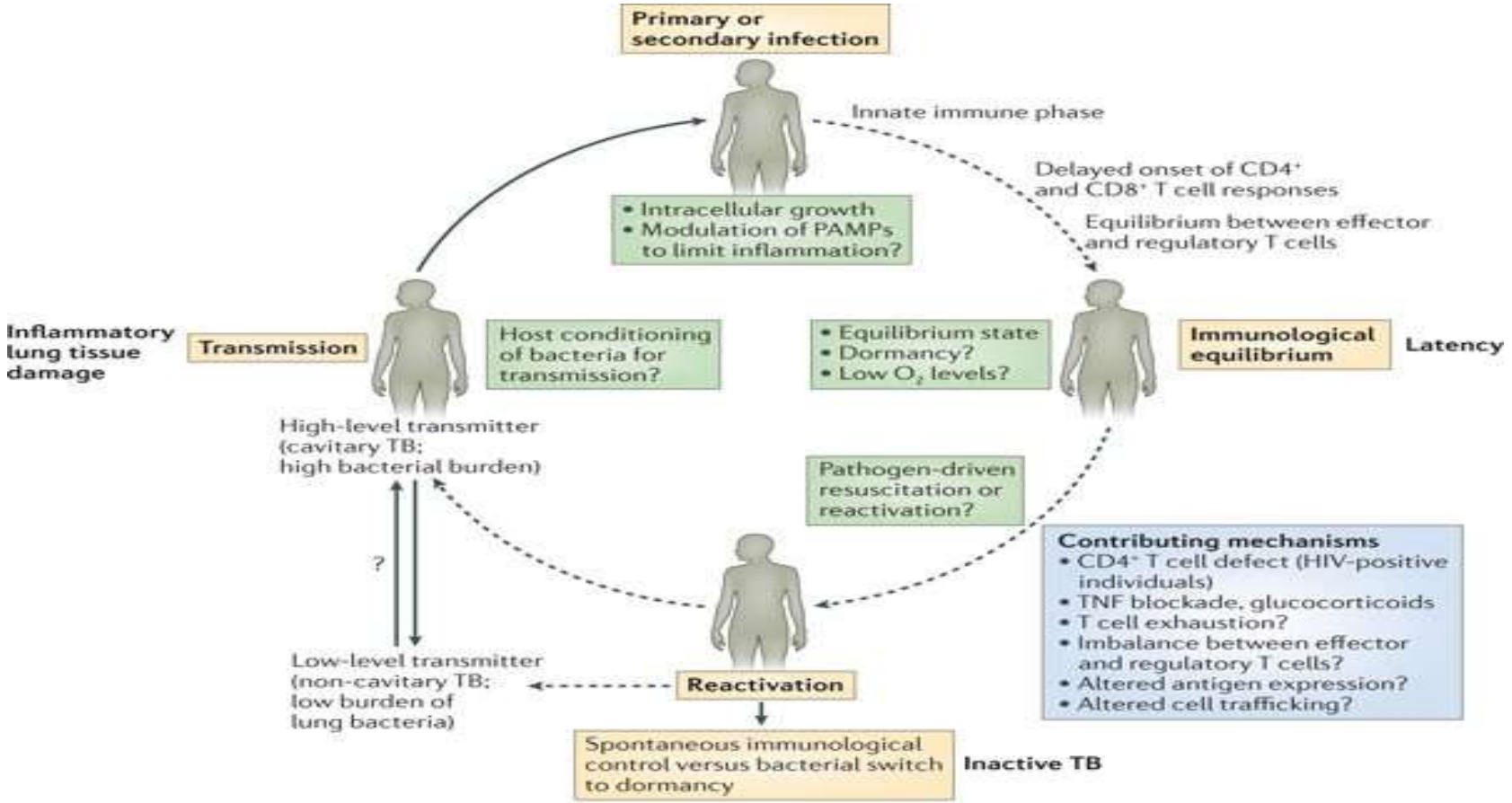
D. Sexually Transmitted Infections (STIs)	E. Central Nervous System (CNS) Infections	c Infections (in immunocom
• **Examples:** HIV/AIDS, Syphilis, Gonorrhea, HPV, Herpes simplex.	• **Examples:** Meningitis (bacterial/viral), Encephalitis, Rabies, Polio.	• HIV/AIDS patients, transplant recipients, chemotherapy patients. • **Examples:**

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Major Infectious Diseases

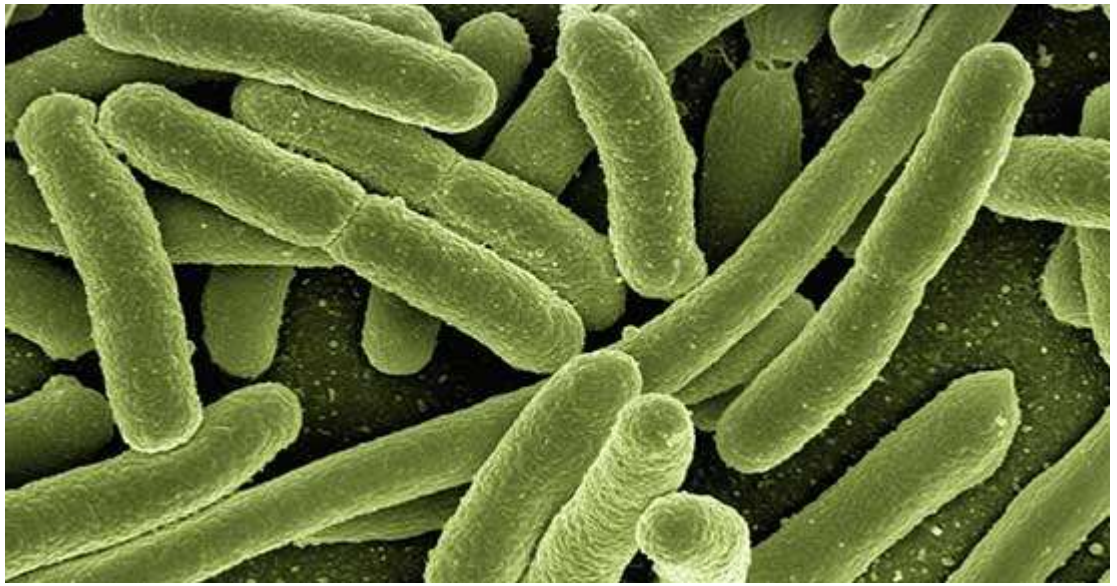
- **Respiratory Infections**
- **Example:** Tuberculosis



Nature Reviews | Immunology

Tuberculosis (TB)

- **Definition**: Tuberculosis (TB) is a chronic infectious disease caused by Mycobacterium tuberculosis, primarily affecting the lungs (pulmonary TB) but can also involve other organs (extrapulmonary TB).
- **Nature**: Airborne communicable disease.
- **Significance**: One of the leading causes of morbidity and mortality worldwide.



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- Main pathogen: ***Mycobacterium tuberculosis*** (MTB) – obligate aerobe, acid-fast bacillus.
- Other species:
  - *M. bovis* – from cattle, zoonotic.
  - *M. africanum* – causes TB in certain regions.
  - *M. microti* and **M. canetti** – rare.

Acid-fast property due to high lipid content in the cell wall (mycolic acids).

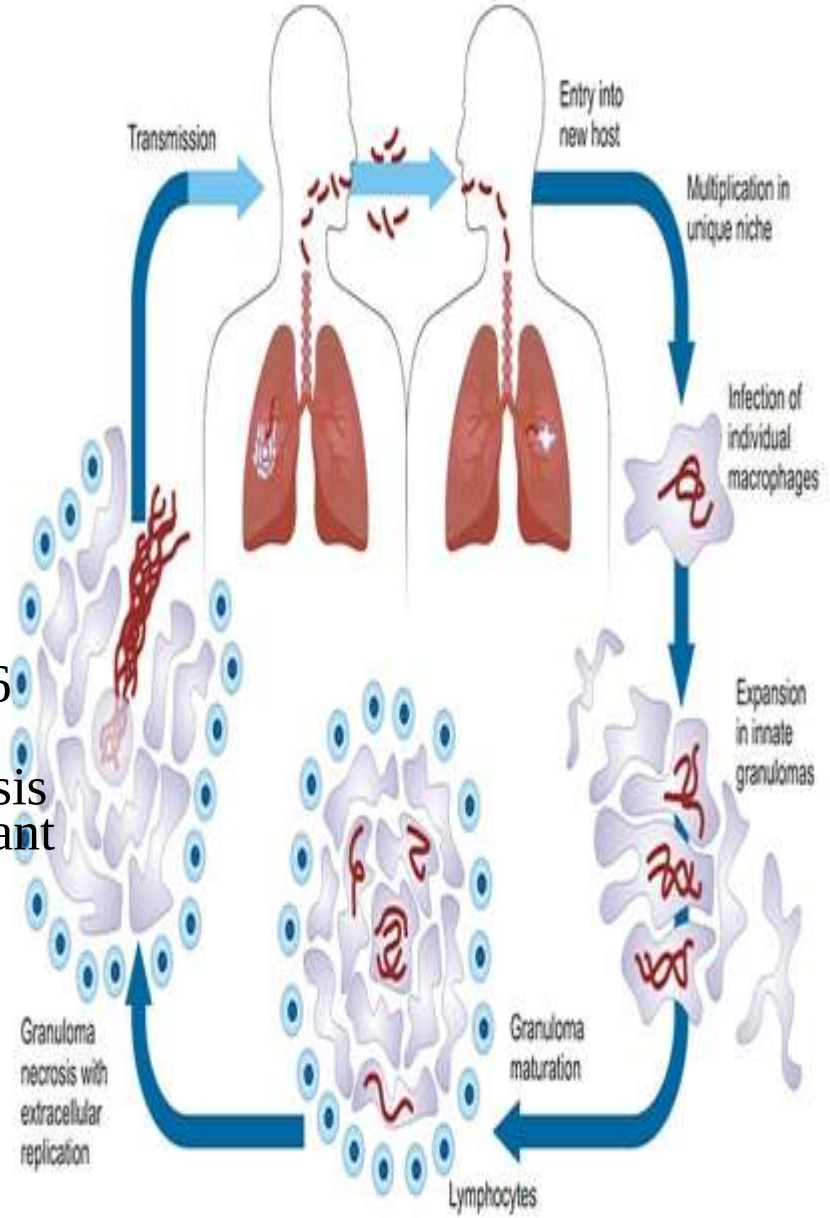
- **Mode of Transmission**
  - **Airborne route**: Inhalation of droplet nuclei (1–5 μm) from coughing, sneezing, or talking by an active TB patient.
  - Less common:
    - Ingestion of unpasteurized milk (M. bovis).
    - Direct inoculation (rare).
  - **Not spread by**: shaking hands, sharing food, touching objects.

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Pathogenesis

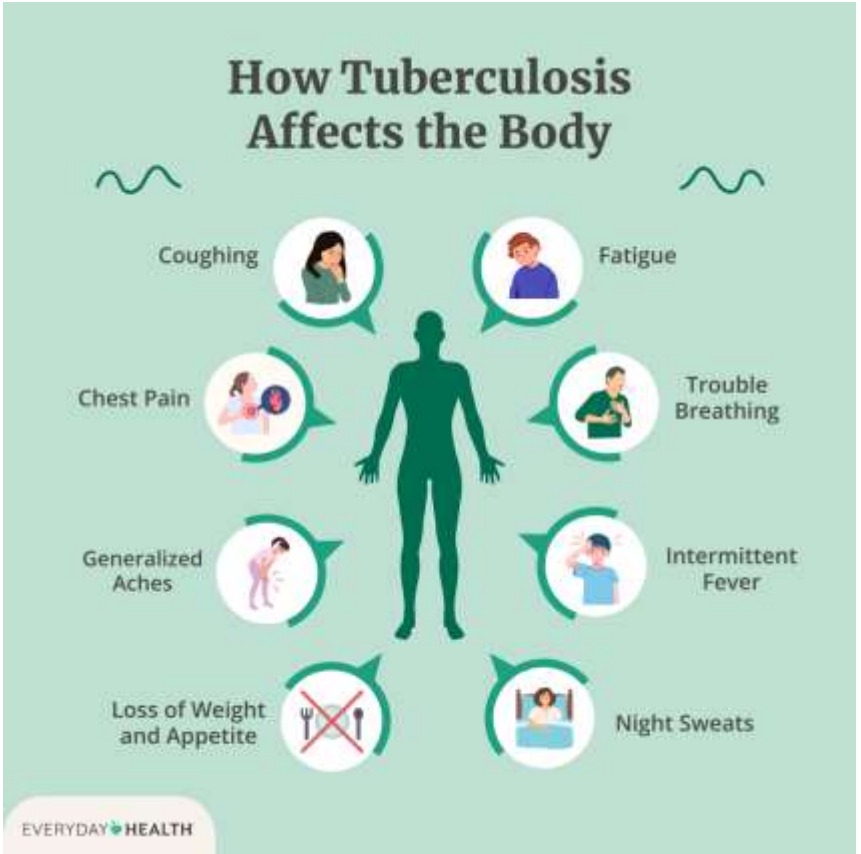
- **Inhalation** of MTB into alveoli.
- **Phagocytosis** by alveolar macrophages.
- MTB resists killing due to:
  - Waxy cell wall (lipids, mycolic acids).
  - Inhibition of phagosome-lysosome fusion.
- **Primary Infection**:
  - Local lesion in lungs (Ghon focus).
  - Spread to hilar lymph nodes → **Primary complex (Ghon complex)**.
- **Immune response**:
  - Cell-mediated immunity develops within 4–6 weeks.
  - Granuloma formation: central caseous necrosis surrounded by epithelioid cells, Langhans giant cells, and lymphocytes.
- **Outcomes**:
  - Healing by fibrosis and calcification.
  - Latent TB infection (dormant bacilli).
  - Progressive disease in immunocompromised hosts.

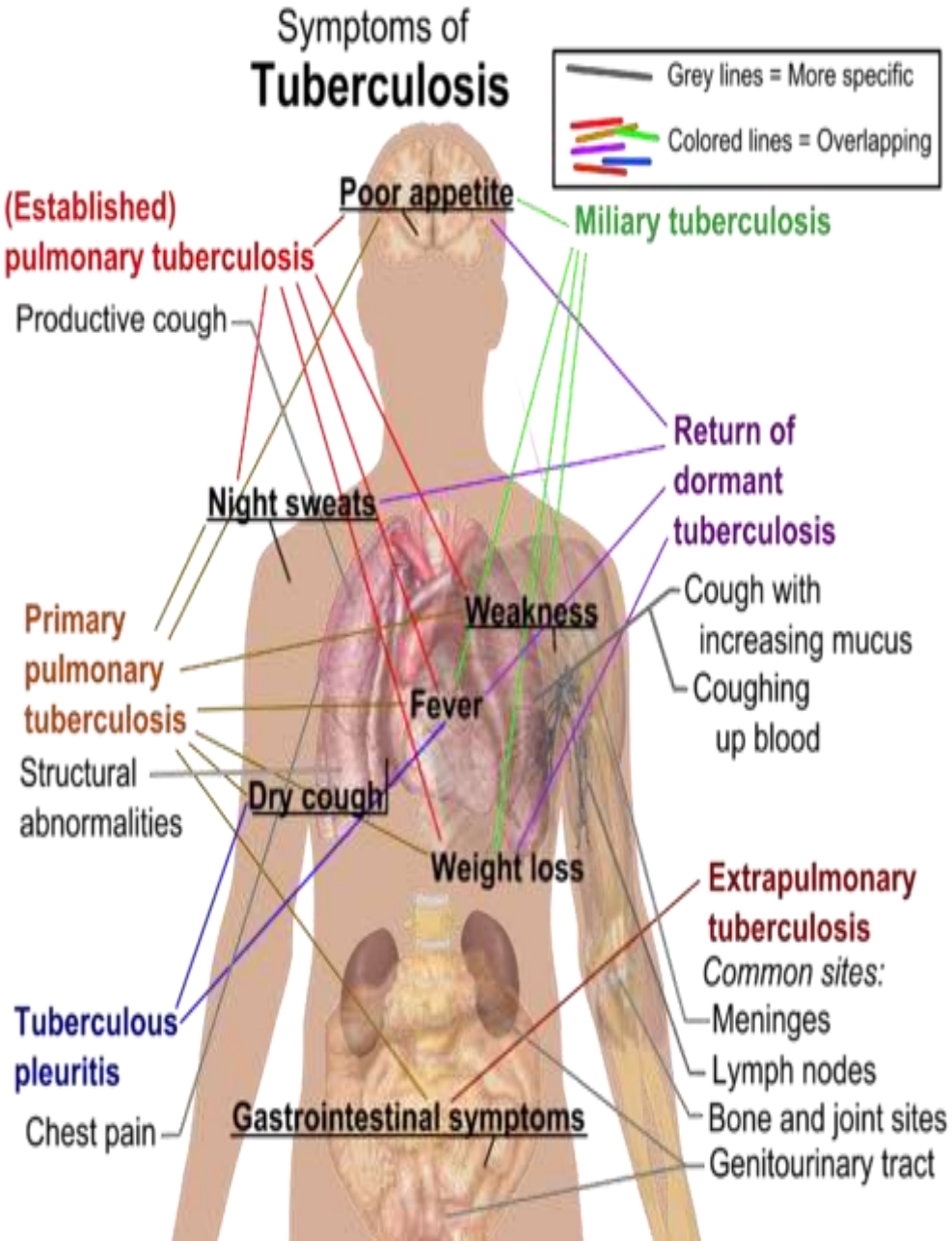


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- **Clinical Features**
  - Persistent cough (>2 weeks).
  - Hemoptysis (blood in sputum).
  - Fever (low grade, evening rise).
  - Night sweats.
  - Weight loss, anorexia, fatigue.
- **Respiratory signs**:
  - Chest pain, breathlessness.
  - Crepitations, bronchial breathing.



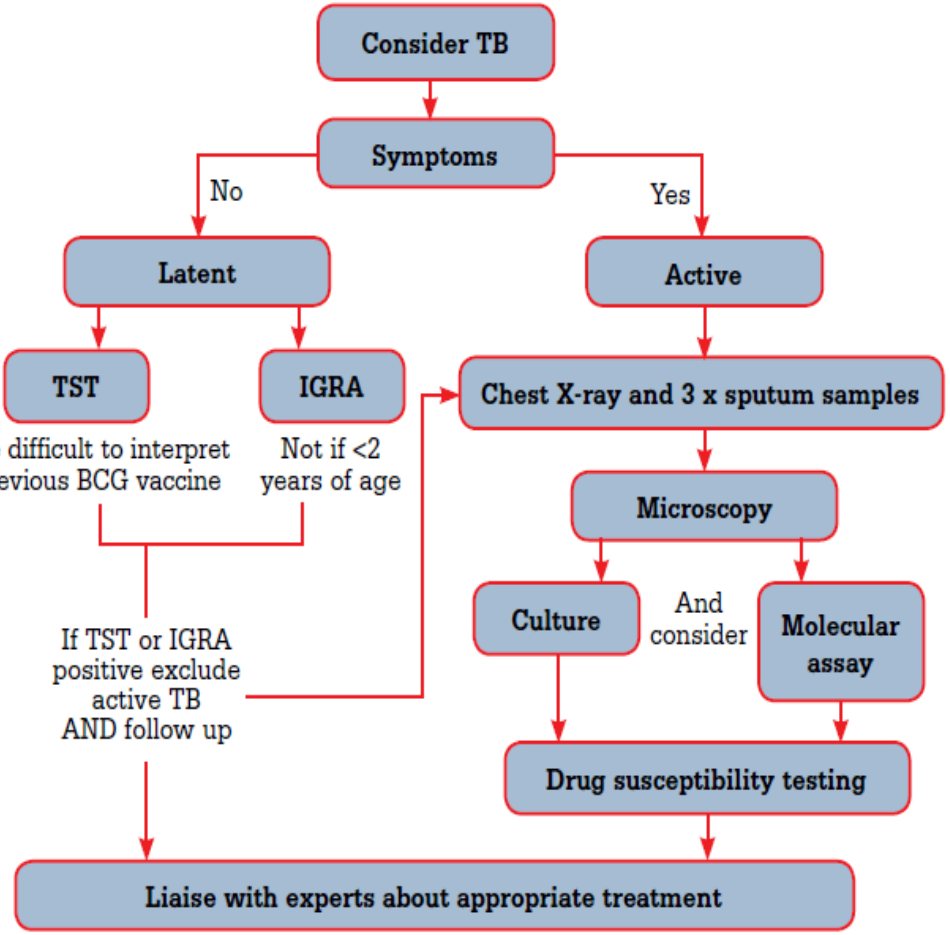


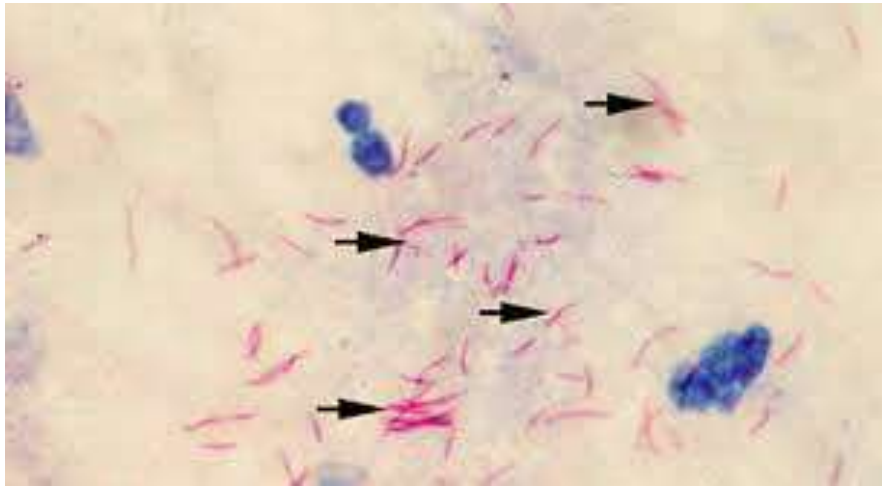
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Diagnosis

- **A. Clinical Suspicion**
- Cough >2 weeks, unexplained fever, weight loss, night sweats.
- **B. Laboratory Investigations**
- **Sputum Microscopy**:
  - Ziehl-Neelsen (ZN) stain → Acid-Fast Bacilli (AFB).
- **Culture** (Gold standard):
  - Lowenstein-Jensen (LJ) medium (takes 4–8 weeks).
  - MGIT (rapid liquid culture).
- **Molecular tests**:
  - GeneXpert/CB-NAAT (detects MTB & rifampicin resistance within 2 hours).
  - PCR-based assays.
- **Tuberculin Skin Test (Mantoux test)**:
  - Positive: ≥10 mm induration after 48–72 hrs (≥5 mm in HIV patients).
- **Interferon Gamma Release Assay (IGRA)**:
  - Measures immune response to MTB antigens.
- **Imaging**:
  - Chest X-ray: cavitary lesions, infiltrates in upper lobes.
  - CT/MRI for extrapulmonary TB.





Treatment

- Long duration, multiple drugs to prevent resistance.
- **First-line drugs (HRZE):**
  - **H**: Isoniazid (INH)
  - **R**: Rifampicin
  - **Z**: Pyrazinamide
  - **E**: Ethambutol
- **Standard Regimen (under NTEP/WHO DOTS):**
  - **Intensive phase**: 2 months (HRZE daily).
  - **Continuation phase**: 4 months (HR + E daily).
  - Total: 6 months (longer for extrapulmonary/complicated TB).
- **Drug-Resistant TB:**
- **MDR-TB**: Resistant to INH + Rifampicin.
- **XDR-TB**: MDR + resistant to fluoroquinolones & injectable drugs.
- Treatment: Second-line drugs (Bedaquiline, Linezolid, Clofazimine, Cycloserine, Fluoroquinolones, etc.).

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Prevention & Control

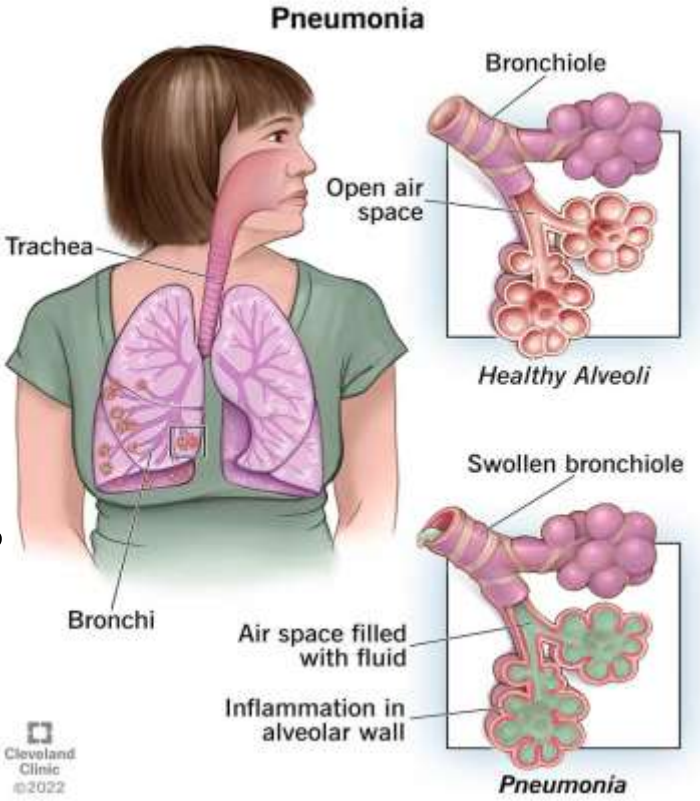
- **BCG vaccine** (Bacillus Calmette-Guérin):
  - Live attenuated *M. bovis* strain.
  - Protects children against severe TB (meningitis, miliary TB).
- **Public health measures**:
  - Early detection & treatment (NTEP in India).
  - Infection control in hospitals.
  - Adequate ventilation, cough hygiene.
  - Contact tracing.
- **National TB Elimination Program (NTEP) – India**
- Aim: **End TB by 2025** (5 years before global WHO target).
- Strategies:
  - Free diagnosis & treatment.
  - GeneXpert as frontline diagnostic.
  - DOTS (Directly Observed Treatment).
  - Nutritional support (Nikshay Poshan Yojana – ₹500/month to patients).
  - Active case finding and contact tracing.

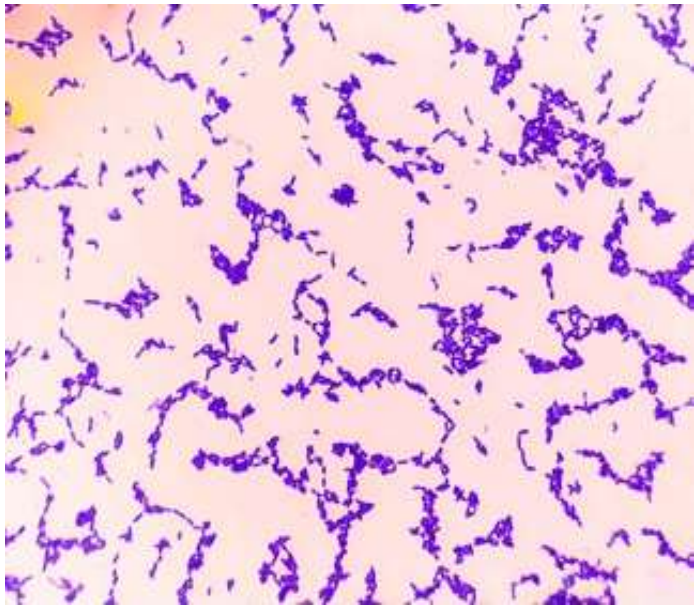
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Pneumonia

- Pneumonia is an **acute infection or inflammation of the lung parenchyma (alveoli and interstitial tissue)** caused by microorganisms (bacteria, viruses, fungi, parasites) or non-infectious agents (aspiration, chemicals).
  It leads to **consolidation of alveoli**, impaired gas exchange, and systemic illness.
- **Causes (Etiology)**
- Pneumonia can be classified based on the causative agents:
- **Bacterial** (most common)
  - *Streptococcus pneumoniae* (typical lobar pneumonia)
  - *Haemophilus influenzae*
  - *Staphylococcus aureus*
  - Gram-negative bacilli: *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, *Escherichia coli*
  - Atypical bacteria: *Mycoplasma pneumoniae*, *Chlamydophila pneumoniae*, *Legionella pneumophila*





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Pathogenesis

- **Entry of pathogen** – via inhalation, aspiration, or hematogenous spread.
- **Colonization** – pathogens bypass host defenses (mucociliary clearance, alveolar macrophages).
- **Inflammatory response** – alveoli fill with exudates, neutrophils, and fibrin (consolidation).
- **Stages of lobar pneumonia (classical pathogenesis):**
  - **Congestion (1–2 days):** Alveoli filled with serous exudate, bacteria, and a few neutrophils.
  - **Red hepatization (2–4 days):** Alveoli filled with RBCs, neutrophils, and fibrin → lung resembles liver.
  - **Gray hepatization (4–8 days):** RBCs break down, fibrin and leukocytes remain.
  - **Resolution (8–10 days):** Macrophages clear debris, lung tissue returns to normal.

Clinical Features

- General symptoms:
  - Fever, chills, fatigue, malaise
  - Loss of appetite, body aches
  - Respiratory symptoms:
  - Cough – productive (purulent or rusty sputum in bacterial), dry (in viral/atypical)
  - Shortness of breath (dyspnea)
  - Chest pain – pleuritic (worsens with breathing/coughing)
  - Tachypnea (rapid breathing),
  - hypoxia (low oxygen saturation)
- Signs on examination:
  - Crackles (crepitations) on auscultation
  - Bronchial breath sounds over consolidated areas
  - Dullness on percussion
  - Increased vocal fremitus
- Complications
  - Pleural effusion (fluid around lungs)
  - Empyema (pus in pleural cavity)
  - Lung abscess
  - Respiratory failure / ARDS (Acute Respiratory Distress Syndrome)
  - Sepsis and septic shock
- Dissemination:
  - Meningitis, endocarditis, pericarditis (esp. with S. aureus)
  - Chronic lung disease (in recurrent pneumonia cases)

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Diagnosis

- **History & Clinical Examination** – symptoms and chest findings.
- **Investigations:**
  - **Chest X-ray:** Lobar consolidation (typical pneumonia), diffuse interstitial pattern (atypical/viral).
  - **Blood tests:** ↑ WBC count (neutrophilia in bacterial, lymphocytosis in viral), ↑ CRP/ESR.
  - **Sputum examination:** Gram stain, culture & sensitivity.
  - **Blood cultures:** to detect septicemia.
  - **Pulse oximetry/ABG:** assess oxygenation.
  - **Special tests:** PCR, antigen detection for viruses/atypical organisms.
  - **Bronchoscopy:** in complicated/refractory cases.

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Treatment & Prevention

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Treatment

- **General supportive care**
  - Bed rest, hydration, nutrition
  - Oxygen therapy if hypoxic
  - Analgesics/antipyretics (paracetamol,

Prevention

- **Vaccination**
  - Pneumococcal vaccines (PCV13, PPSV23)
  - Influenza vaccine (annual)
  - COVID-19 vaccination
- **General preventive measures**

Circulatory Disorders

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- Circulatory disorders are a group of pathological conditions in which the **normal blood flow and distribution of oxygen, nutrients, and waste products** are disturbed due to abnormalities in the blood, blood vessels, or heart.
- They can be **local** (restricted to a part of the body) or **systemic** (involving the whole circulation).

Types of Circulatory Disorders

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1. Hyperemia and Congestion

- **Hyperemia:** Active increase in blood flow due to arteriolar dilation.
  - *Causes:* Exercise, inflammation, blushing.
  - *Features:* Tissue appears red and warm.
- **Congestion:** Passive accumulation of blood due to impaired venous outflow.
  - *Causes:* Heart failure, venous obstruction.
  - *Features:* Tissue appears cyanotic (blue-purple).

Types of Circulatory Disorders

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2. Hemorrhage

- **Definition:** Escape of blood from vessels due to rupture or damage.
- **Types:**
  - *External hemorrhage* – visible outside the body.
  - *Internal hemorrhage* – within tissues or body cavities (e.g., brain, peritoneum).
- **Clinical examples:** Epistaxis (nosebleed), hemoptysis (coughing blood), hematemesis (vomiting blood), cerebral hemorrhage.
- **Complication:** Shock, anemia, death if severe.

Types of Circulatory Disorders

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3. Thrombosis

- **Definition:** Formation of a solid mass (thrombus) from blood constituents inside intact vessels or heart.
- **Causes (Virchow's Triad):**
  - Endothelial injury (trauma, atherosclerosis).
  - Abnormal blood flow (stasis or turbulence).
  - Hypercoagulability of blood (genetic, malignancy, dehydration).
- **Consequences:**
  - Occlusion of blood vessel → ischemia/infarction.
  - Embolization → thromboembolism.

Types of Circulatory Disorders

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4. Embolism

- **Definition:** Passage of abnormal material (embolus) through circulation causing vessel blockage.
- **Types of emboli:**
  - **Thromboembolism** (most common – detached thrombus).
  - **Fat embolism** (after fractures of long bones).
  - **Air embolism** (decompression sickness, IV line accidents).
  - **Amniotic fluid embolism** (during childbirth).
  - **Septic embolism** (infected material from endocarditis).
- **Effects:** Pulmonary embolism, stroke, organ infarction.

Types of Circulatory Disorders

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5. Infarction

- **Definition:** Localized area of ischemic necrosis caused by obstruction of blood supply.
- **Causes:** Thrombosis, embolism, arterial spasm, torsion/strangulation of organs.
- **Types:**
  - **White (anemic) infarct** – in solid organs with end-arterial supply (heart, kidney, spleen).
  - **Red (hemorrhagic) infarct** – in lungs, intestine (with dual blood supply or venous obstruction).
- **Complications:** Scar formation, organ dysfunction, heart failure (myocardial infarction).

Types of Circulatory Disorders

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6. Shock

- **Definition:** Acute circulatory failure with inadequate tissue perfusion and oxygenation.
- **Types:**
  - **Hypovolemic shock** – due to blood or fluid loss (hemorrhage, burns, vomiting, diarrhea).
  - **Cardiogenic shock** – due to pump failure (MI, arrhythmia, myocarditis).
  - **Septic shock** – due to severe infection, endotoxins cause vasodilation & capillary leakage.
  - **Anaphylactic shock** – due to severe allergic reaction (histamine release → vasodilation, bronchospasm).
  - **Neurogenic shock** – due to spinal cord injury or anesthesia (loss of sympathetic tone).
- **Clinical features:** Hypotension, cold clammy skin (except septic shock – warm skin), rapid weak pulse, altered sensorium, oliguria.
- **Complications:** Multi-organ failure, death.

Types of Circulatory Disorders

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7. Edema

- **Definition:** Abnormal accumulation of fluid in interstitial tissue spaces.
- **Causes (Pathophysiology):**
  - ↑ Hydrostatic pressure (heart failure, venous obstruction).
  - ↓ Plasma oncotic pressure (hypoproteinemia in nephrotic syndrome, liver disease, malnutrition).
  - Lymphatic obstruction (tumors, filariasis).
  - Increased capillary permeability (inflammation, allergy).
- **Types:**
  - Localized (e.g., limb edema in DVT).
  - Generalized (anasarca – severe hypoproteinemia).
- **Complications:** Impaired healing, infections, breathing difficulty in pulmonary edema.

Types of Circulatory Disorders

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8. Disseminated Intravascular Coagulation (DIC)

- **Definition:** Pathological activation of coagulation cascade leading to widespread microthrombi and consumption of clotting factors → paradoxical bleeding.
- **Causes:** Sepsis, trauma, malignancy, obstetric complications.
- **Features:** Bleeding, organ ischemia, shock.
- **Complications:** Multi-organ failure, death.

Diagnosis, Treatment & Prevention

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Diagnosis	Treatment	Prevention
• Circulatory disorders are diagnosed using a combination of: • **History and examination** (swelling,	• Management depends on the specific disorder: • **Hemorrhage:** Control bleeding, blood transfusion,	• Regular exercise and a healthy diet (to prevent atherosclerosis and thrombosis). • Avoid

Symptoms and signs related to diseases of the various body systems

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1. Cardiovascular System

- The cardiovascular system includes the heart and blood vessels. Diseases here often affect blood circulation, leading to characteristic features.

- **Common Symptoms**
  - **Chest pain or discomfort** → seen in angina, heart attack. Patients may describe it as heaviness, squeezing, or burning.
  - **Palpitations** → awareness of one's heartbeat, may feel fast, irregular, or pounding.
  - **Shortness of breath (dyspnea)** → due to poor pumping of the heart or fluid in lungs.
  - **Swelling (edema)** → commonly in legs and ankles, due to fluid retention in heart failure.
  - **Fatigue and weakness** → reduced blood supply to tissues.
- **Common Signs**
  - Irregular pulse, murmurs, raised jugular venous pressure, bluish discoloration (cyanosis), enlarged heart on examination.

Symptoms and signs related to diseases of the various body systems

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- **2. Respiratory System**
- This system involves the lungs and airways. Any disorder here interferes with breathing and oxygen supply.
- **Symptoms**
  - **Cough** → dry or productive (with sputum), important in infections like bronchitis, pneumonia.
  - **Breathlessness** → in asthma, COPD, lung infections, pulmonary embolism.
  - **Chest pain** → sharp pain that worsens on breathing may indicate pleurisy.
  - **Wheezing** → high-pitched whistling sound, common in asthma.
  - **Hemoptysis** → coughing up blood, seen in tuberculosis or lung cancer.
- **Signs**
  - Abnormal breath sounds (crackles, wheezes), clubbing of fingers, cyanosis, rapid breathing.

Symptoms and signs related to diseases of the various body systems

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- **3. Gastrointestinal (GI) System**
- Covers mouth, stomach, intestines, liver, and pancreas. Problems here affect digestion and absorption.
- **Symptoms**
  - **Abdominal pain** → may be localized (appendicitis) or diffuse (gastroenteritis).
  - **Nausea and vomiting** → common in gastritis, food poisoning, obstruction.
  - **Diarrhea** → frequent watery stools, may indicate infection or malabsorption.
  - **Constipation** → difficulty in passing stools.
  - **Heartburn** → burning behind the breastbone due to acid reflux.
  - **Jaundice** → yellowing of eyes and skin in liver disease.
- **Signs**
  - Abdominal tenderness, enlarged liver or spleen, visible distension, abnormal bowel sounds.

Symptoms and signs related to diseases of the various body systems

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- **4. Nervous System**
- Includes the brain, spinal cord, and nerves. Disorders can be very disabling.
- **Symptoms**
  - **Headache** → in migraine, meningitis, brain tumor.
  - **Dizziness or vertigo** → imbalance or spinning sensation.
  - **Seizures (fits)** → sudden episodes of abnormal movement or behavior.
  - **Numbness, tingling, weakness** → seen in nerve damage or stroke.
  - **Memory loss, confusion** → in dementia, delirium.
- **Signs**
  - Weakness or paralysis of limbs, abnormal reflexes, tremors, slurred speech, loss of coordination, altered consciousness.

Symptoms and signs related to diseases of the various body systems

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- **5. Genitourinary System**
- Includes kidneys, bladder, and reproductive organs.
- **Symptoms**
  - **Painful urination (dysuria)** → urinary tract infection.
  - **Frequent urination** → diabetes, infections.
  - **Blood in urine (hematuria)** → stones, infection, tumors.
  - **Flank pain** → kidney stones or infection.
  - **Incontinence** → inability to control urination.
- **Signs**
  - Tenderness over kidney area, enlarged prostate in men, swelling due to kidney failure.

Symptoms and signs related to diseases of the various body systems

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- **6. Endocrine System**
- This system controls hormones, and diseases often affect metabolism.
- **Symptoms**
  - **Excessive thirst and urination** → diabetes mellitus.
  - **Weight changes** → weight loss in hyperthyroidism, weight gain in hypothyroidism or Cushing's.
  - **Heat or cold intolerance** → thyroid disorders.
  - **Fatigue, lethargy** → adrenal or thyroid dysfunction.
- **Signs**
  - Goiter (enlarged thyroid), abnormal facial features (acromegaly, Cushing's), pigmentation changes, delayed growth in children.

Symptoms and signs related to diseases of the various body systems

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- **7. Musculoskeletal System**
- Covers bones, joints, muscles.
- **Symptoms**
  - **Joint pain (arthralgia)** → arthritis, gout.
  - **Stiffness** → especially in morning (rheumatoid arthritis).
  - **Back pain** → disc problems, strain.
  - **Muscle weakness** → myopathies.
- **Signs**
  - Swollen joints, deformities, reduced movement, muscle wasting, tenderness.

Symptoms and signs related to diseases of the various body systems

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- **8. Skin and Integumentary System**
- Skin reflects internal and external health issues.
- **Symptoms**
  - **Itching (pruritus)** → allergies, infections, liver disease.
  - **Rashes** → in infections, autoimmune conditions.
  - **Painful ulcers or blisters** → herpes, burns.
- **Signs**
  - Color changes (pallor, cyanosis, jaundice), dryness, scaling, lesions, nail changes.

Symptoms and signs related to diseases of the various body systems

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- **9. Hematological System (Blood Disorders)**
- **Symptoms**
  - Fatigue, weakness → anemia.
  - Easy bruising or bleeding → clotting disorders.
  - Recurrent infections → low white cell count.
- **Signs**
  - Pallor, enlarged lymph nodes, petechiae (small red spots), enlarged spleen.

Symptoms and Signs Related to Diseases of Control Systems

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- The body maintains stability through three key control systems:
- **Nervous system** → fast, electrical signaling
- **Endocrine system** → slower, hormonal control
- **Immune system** → defense, recognition, and regulation
- When these systems malfunction, characteristic **symptoms (patient-reported experiences)** and **signs (clinically observable features)** appear.

1. Nervous System Disorders

The nervous system controls movement, sensation, thinking, and coordination.

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A. Symptoms

- **Headache**:
  - Throbbing in migraine, generalized in meningitis, sudden and severe in subarachnoid hemorrhage.
- **Dizziness / Vertigo**:

B. Signs

- **Motor abnormalities**: tremors (Parkinson's), rigidity, abnormal postures.
- **Gait changes**: shuffling (Parkinson's), ataxic gait (cerebellar

2. Endocrine System Disorders

This system regulates metabolism, growth, reproduction, and stress. Disorders arise from **excess, deficiency, or resistance** to hormones.

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A. Symptoms

- **Thyroid disorders**:
  - Hyperthyroidism → weight loss, heat intolerance, sweating, tremors, irritability.
  - Hypothyroidism → weight gain, cold intolerance

B. Signs

- **Visible changes**:
  - Goiter (enlarged thyroid), exophthalmos (protruding eyes in Graves' disease).
  - Obesity with "moon face," "buffalo hump."

3. Immune System Disorders

This system recognizes and destroys pathogens but can overreact or turn against the body.

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A. Symptoms

- **Recurrent infections**: suggest immunodeficiency (HIV, congenital).
- **Unusual fatigue**: chronic inflammatory and autoimmune diseases.

B. Signs

- **Physical examination**:
  - Enlarged lymph nodes (lymphadenopathy).
  - Splenomegaly (enlarged spleen).
  - Oral thrush,